

Devi AI - Technology Stack Summary

Complete Technology Stack & Project Overview

Project Overview

Devi is a production-grade, AI-powered DevOps dashboard and operating system. It provides a unified web interface to manage infrastructure through natural language commands, voice control, and traditional UI controls. The system combines real-time monitoring, container orchestration, Kubernetes management, CI/CD pipelines, and an intelligent AI agent into a single platform.

Core Features:

- AI Agent - Natural language interface for infrastructure management (Jarvis-style)
- Docker Management - Monitor, create, start/stop/restart containers, build images
- Kubernetes Dashboard - Pod, node, deployment management with scaling & rollbacks
- CI/CD Pipeline - Track deployments, pipelines, history, and rollbacks
- Real-time Monitoring - CPU, memory, disk, network metrics with live WebSocket updates
- In-browser Terminal - Command execution with safety blocklist/allowlist controls
- Voice Assistant - Wake-word activated voice control using Web Speech API
- Activity Feed - Real-time infrastructure event stream
- Authentication - JWT with refresh tokens, role-based access (admin, operator, viewer)
- API Documentation - OpenAPI 3.0 / Swagger UI at /api/docs
- Anomaly Detection - AI agent detects CPU spikes, memory leaks, restart loops

Architecture:

The system follows a client-server architecture. The Next.js frontend communicates with the Express backend via REST APIs (axios) and WebSockets (socket.io). The backend orchestrates infrastructure through Docker Engine API (dockerode), Kubernetes API (@kubernetes/client-node), and system metrics (systeminformation). The AI agent processes natural language through a rule-based intent engine with session memory.

Default Credentials: admin / admin123

Project Structure: dev/ (Next.js frontend), dev/backend/ (Express API), dev-hero/ (Vite landing page), deploy/ (Docker Compose + Nginx + scripts)

1. Frontend - Next.js Application (dev/)

Next.js	16.2.6	App Router, React Server Components
React	19.2.4	Client components with 'use client'
TypeScript	^5	Strict mode, ES2017 target
Tailwind CSS	^4	Utility-first CSS framework
@tailwindcss/postcss	^4	Tailwind PostCSS plugin
axios	^1.16.1	HTTP client, pre-configured interceptors
framer-motion	^12.39.0	Page/component animations
lucide-react	^1.16.0	SVG icon library
recharts	^3.8.1	Charts for system metrics

Devi AI - Technology Stack Summary

socket.io-client	^4.8.3	Real-time WebSocket client
eslint	^9	Flat config linter
eslint-config-next	16.2.6	Next.js ESLint rules
@types/node	^20	Node type definitions
@types/react	^19	React type definitions
@types/react-dom	^19	ReactDOM type definitions

Routes: / (home), /login, /dashboard, /agent, /chat, /docker, /kubernetes, /cicd, /monitoring, /terminal, /logs, /settings, /deploy

Components: ActivityFeed, AICore, AppShell, AuthGuard, BootScreen, GlassCard, HolographicCard, PageTransition, Particles, Sidebar, Toast, VoiceIndicator

Hooks: useVoiceAssistant (Web Speech API)

Library: api.ts (Axios), auth.tsx (JWT Context), socket.ts (Socket.io), utils.ts (cn, formatBytes, formatUptime)

Fonts: Geist (display), Inter (body), JetBrains Mono (code), Material Symbols (icons)

2. Backend - Express API Server (devi/backend/)

Node.js	22-alpine	Docker runtime base
TypeScript	^5.6	ES2022 target, bundler resolution
Express	^4.21.0	Web framework
tsx	^4.19.0	TypeScript dev runner (watch mode)
socket.io	^4.8.0	WebSocket server
bcryptjs	^3.0.3	Password hashing (10 rounds)
jsonwebtoken	^9.0.3	JWT auth (24h expiry)
cors	^2.8.5	CORS with origin whitelist
helmet	^8.0.0	Security headers
morgan	^1.10.0	HTTP request logging
express-rate-limit	^8.5.2	120 req/min rate limiting
dockerode	^5.0.0	Docker Engine API client
@kubernetes/client-node	^1.4.0	K8s API (pods, nodes, deployments)
systeminformation	^5.31.6	System metrics (CPU, RAM, disk)
swagger-jsdoc	^6.2.8	OpenAPI 3.0 spec generation
swagger-ui-express	^5.0.1	Swagger UI at /api/docs
tar	^7.5.15	Tar streams for Docker builds
dotenv	^16.4.5	Environment config

Services: system.ts, docker.ts, kubernetes.ts, auth.ts, ai.ts, agent.ts, parser.ts, shell.ts, safety.ts, memory.ts, logs.ts, cicd.ts

API Routes: /api/auth, /api/system, /api/docker, /api/kubernetes, /api/logs, /api/ai, /api/terminal, /api/cicd, /api/agent, /api/deploy

WebSocket Events: system:update, activity:new, logs:line, terminal:input/output, deployment:*

Middleware: helmet, cors, morgan, express.json, rate-limit, auth/requireAuth, auth/requireRole

3. Landing Page - Vite React (devi-hero/)

Devi AI - Technology Stack Summary

React	^19.2.6	UI library
TypeScript	~6.0.2	Latest TypeScript
Vite	^8.0.12	Build tool & dev server
@vitejs/plugin-react	^6.0.1	React Fast Refresh
@tailwindcss/vite	^4.3.0	Tailwind Vite plugin
tailwindcss	^4.3.0	Utility-first CSS
lucide-react	^1.16.0	SVG icons
eslint	^10.3.0	Linter with flat config
typescript-eslint	^8.59.2	TypeScript ESLint

4. Design Assets (extracted/)

Deep Space HUD design system with cinematic glassmorphism theme.

Color: Dark (#020617), Cyan (#8aebff), Purple (#ddb7ff), Lime (#d9e70d)

Typography: Geist, Inter, JetBrains Mono | Layout: 12-col grid, 1440px max

Mockups: AI Assistant, DevOps Dashboard, Docker Management, Landing Page

5. Infrastructure & DevOps Tooling

Docker Compose	v3.8	Frontend + Backend orchestration
Dockerfile	multi-stage	node:20-alpine with docker-cli
Nginx	reverse-proxy	WebSocket support, gzip, rate limiting
Deploy Scripts	PowerShell	deploy.ps1 lifecycle management
Health Checks	curl + Docker	HEALTHCHECK on both services
Networking	bridge	devi_net network, devi_data volume
Ports	3000 / 4000	Frontend / Backend exposure

6. Complete Technology Inventory (Alphabetical)

Technology	Project	Version	Purpose
@eslint/js	devi-hero	^10.0.1	ESLint JS rules
@kubernetes/client-node	backend	^1.4.0	Kubernetes API client
@tailwindcss/postcss	devi	^4	Tailwind PostCSS plugin
@tailwindcss/vite	devi-hero	^4.3.0	Tailwind Vite plugin
@vitejs/plugin-react	devi-hero	^6.0.1	React Fast Refresh
axios	devi	^1.16.1	HTTP client
bcryptjs	backend	^3.0.3	Password hashing
cors	backend	^2.8.5	CORS middleware
dockerode	backend	^5.0.0	Docker Engine API
dotenv	backend	^16.4.5	Environment config
eslint	devi	^9	Linter
eslint	devi-hero	^10.3.0	Linter

Devi AI - Technology Stack Summary

Technology	Project	Version	Purpose
eslint-config-next	devi	16.2.6	Next.js ESLint rules
eslint-plugin-react-hooks	devi-hero	^7.1.1	React Hooks rules
eslint-plugin-react-refresh	devi-hero	^0.5.2	React Refresh rules
express	backend	^4.21.0	Web framework
express-rate-limit	backend	^8.5.2	Rate limiting
framer-motion	devi	^12.39.0	Animations
globals	devi-hero	^17.6.0	ESLint globals
helmet	backend	^8.0.0	Security headers
jsonwebtoken	backend	^9.0.3	JWT auth
lucide-react	devi	^1.16.0	Icons
lucide-react	devi-hero	^1.16.0	Icons
morgan	backend	^1.10.0	Request logging
next	devi	16.2.6	React framework
react	devi	19.2.4	UI library
react	devi-hero	^19.2.6	UI library
react-dom	devi	19.2.4	React DOM
react-dom	devi-hero	^19.2.6	React DOM
recharts	devi	^3.8.1	Charting
socket.io	backend	^4.8.0	WebSocket server
socket.io-client	devi	^4.8.3	WebSocket client
swagger-jsdoc	backend	^6.2.8	OpenAPI spec
swagger-ui-express	backend	^5.0.1	API docs UI
systeminformation	backend	^5.31.6	System metrics
tailwindcss	devi	^4	CSS framework
tailwindcss	devi-hero	^4.3.0	CSS framework
tar	backend	^7.5.15	Tar streams
tsx	backend	^4.19.0	TS dev runner
typescript	devi	^5	Type system
typescript	backend	^5.6.0	Type system
typescript	devi-hero	~6.0.2	Type system
typescript-eslint	devi-hero	^8.59.2	TS ESLint
vite	devi-hero	^8.0.12	Build tool

7. Non-Package Technologies

Technology	Project	Purpose
Web Speech API	devi frontend	Voice assistant (wake word, STT)
Docker CLI (dockerode)	backend	Container management
Kubernetes (kubernetes)	backend	Cluster management
Nginx	deploy/	Reverse proxy, WebSocket, gzip
OpenAPI 3.0 / Swagger	backend	API documentation
Docker Compose v3.8	root	Service orchestration
PowerShell / Batch	deploy/	Deployment scripts
Google Fonts	devi	Geist, Inter, JetBrains Mono
CSS Animations	devi	shimmer, pulse, float, scan

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Devi AI - AI-Powered DevOps Infrastructure